Chapter 8
Change in Occupied Wolf Habitat in the Northern Great Lakes Region

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8.1 Introduction

The concept of wolf habitat and relative suitability has changed significantly over the past several decades. In large part, this occurred because of insights gained during expansion of the wolf population in the northern Great Lakes states (Mech 1970; Erb and DonCarlos, this volume; Beyer et al., this volume; Wydeven et al., this volume). Protection from intentional killing of wolves since 1974, under the Endangered Species Act of 1973, began the process of wolf population growth and expansion in northeastern Minnesota, with eventual recolonization of northern Wisconsin and upper Michigan (Beyer et al., this volume; Wydeven et al., this volume).

In 1955, Wisconsin game manager John Keener wrote about wolves, "This animal is a symbol of the true wilderness. He cannot tolerate the advancing civilization of his wild home" (Keener 1955: 22). As late as the 1980s it was still generally believed that wolves required wilderness to survive, though research was beginning to show otherwise (Mech et al. 1988; Mech 1989). This concept lasted for so long in part because wolves had persisted only in the Boundary Waters Canoe Area Wilderness and adjacent areas of the Superior National Forest in northeastern Minnesota (Erb and DonCarlos, this volume), as well as Isle Royale National Park in Lake Superior (Vucetich and Peterson, this volume). Gradually, it became clearer that the role of wilderness was largely one of protection for wolves from killing through reduced human accessibility, rather than any innate requirements of wolves and their behavior (Mech 1995).

With protection, wolves colonized areas with greater human presence. At the same time, remoteness clearly has a positive effect on wolf survival because of reduced conflict with humans, reduced accidental killing of wolves (such as by vehicles), and perhaps less disease, as well as less intentional illegal killing. Remoteness provides one relative factor in defining degrees of habitat suitability for wolves. The other important factor is prey abundance. Ironically, in today's human-dominated landscape, these factors are often in conflict. Human-dominated landscapes, both forests subject to harvesting and re-growth and agricultural lands, support high levels of prey (white-tailed deer, *Odocoileus virginianus*) abundance.

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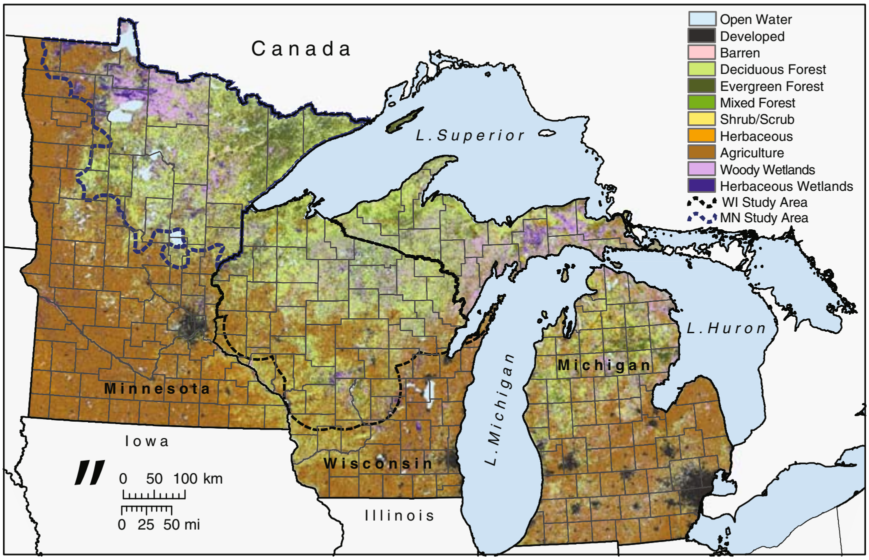

Fig. 8.1 Current land cover in Minnesota, Wisconsin, and Michigan. Data source: USGS National Land Cover Database 2001 (2006). Map created: Forest Landscape Ecology Laboratory, Department of Forest and Wildlife Ecology, University of Wisconsin-Madison. © David J. Mladenoff (*see Color Plate 1*)

But by supporting higher abundance of prey, these human-dominated landscapes can also support a higher wolf population. Greater wolf–human conflict ultimately often results (Mladenoff et al. 1997).

Areas with more intensive forest management and agriculture provide highly productive deer habitat. In the Great Lakes states, this gradient of productivity is controlled primarily by climate and soils, and declines generally from north to south. This gradient today largely follows the increasing area and productivity of agriculture from north to south (Fig. 8.1). Agriculture maintains productive browse and grazing within reach of deer, and the intensive nutrient inputs to crops and pasture accentuate the attractiveness of crops to deer, as well as providing livestock (prey) availability to wolves (Treves et al. 2002).

8.2 Historical Vegetation and Habitat

Prior to widespread Euro-American settlement in the 1800s, all ecosystems in the region were wolf habitat. Indeed, this is the case nearly worldwide. Wolves occur, or occurred, wherever there was prey (Mech 1995).

For a top predator like the wolf, the most important component of habitat is adequate prey, and for wolves the major prey species are ungulates. Where prairie and open savanna occurred, bison (*Bison bison*) and elk (*Cervus elaphus*) would have been most abundant. Furthest north, woodland caribou (*Rangifer tarandus*) and moose (*Alces alces*) were most abundant. In the mixed conifer-deciduous forests in between, white-tailed deer were most abundant (DelGiudice et al., this volume). These general relationships were similar even as vegetation zones oscillated with climate over recent millennia. Prey species and their relative abundance varied as influenced by climate and productivity, as well as hunting by Native Americans.

Our best picture of the distribution of regional ecosystems before modern levels of change is from the mid- to late 1800s, the period of the US General Land Office Survey. While the main purpose of this survey was to facilitate sale of the public domain and settlement, surveyors also recorded valuable information on the natural systems present. Also, because of the land survey, this additional information is well referenced spatially, and can be transcribed and mapped (Schulte and Mladenoff 2005). To varying degrees, this occurred in the three states of the Great Lakes region. More detailed state efforts (Schulte et al. 2002) generalize to a regional picture of ecosystems present before the main influx of Euro-American settlers, logging, agriculture, and wolf extermination (Fig. 8.2).

The Great Lakes region was dominated by prairie and oak (*Quercus*) savanna to the west and south, mixed evergreen-deciduous forest in northern Wisconsin

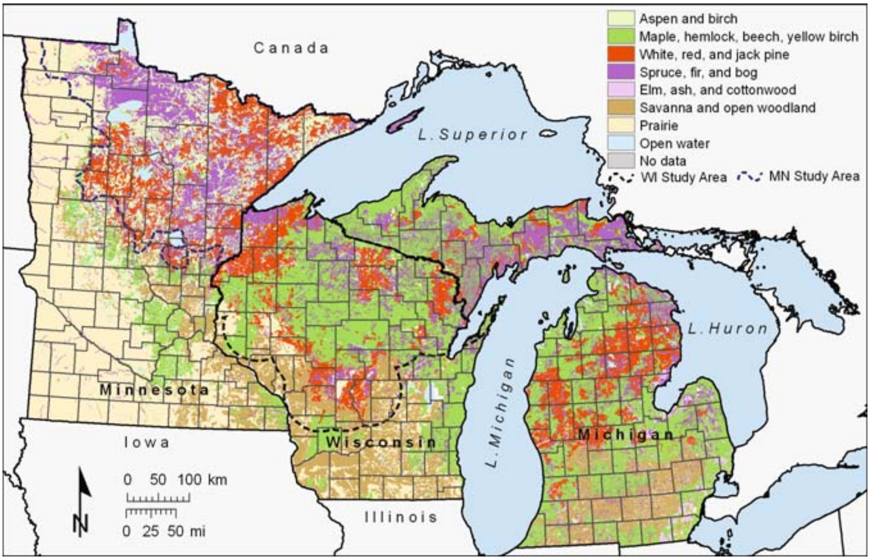

Fig. 8.2 Pre-European land cover (1800s) in the Great Lakes region. Data compiled for the three states from the US Government Land Office survey of the 1800s. Data source: US Forest Service Great Lakes Assessment. Land cover classification and map: Forest Landscape Ecology Laboratory, Department of Forest and Wildlife Ecology, University of Wisconsin-Madison. © David J. Mladenoff (*see Color Plate 2*)

and upper Michigan on heavier moraine soils, and pine (*Pinus*) on sandy outwash plains (Curtis 1959). Northeastern Minnesota was distinctive with a mixture of pine- and aspen- (*Populus*) dominated uplands, and abundant spruce-fir (*Picea Abies*) in lowlands, and in some uplands (Heinselman 1996). The possible patterns of ungulate and wolf abundance that occurred over time were unclear and potentially complex. They also would have been dynamic. Ungulate, and therefore wolf, biomass generally should have followed patterns of vegetation productivity, which because of climate (higher temperature and precipitation) and better soils, would have been most abundant in the southeastern Great Lakes states region. However, to the east, increasing vegetation productivity went largely into woody plant tissue (trees). In the western portion of the region in the prairie and savannas, also very productive on good soils, much above-ground plant productivity was within the reach of ungulates, but only seasonally. Also, the higher productivity in the southeast would have been tempered by a similarly higher Native American population and their impact on ungulate populations (Cleland 1983). Native American agriculture also was more important in the southeastern Great Lakes region and therefore the net level of competition for prey with humans is not obvious. Competition with Native Americans for ungulate prey may have been greater to the north, where hunter-gatherer cultures were more prevalent (Cleland 1983).

In the northern Great Lakes region, the most productive soils were dominated by mixed forest, and in particular in northern Wisconsin and upper Michigan closed canopy forests of sugar maple (*Acer saccharum*), hemlock (*Tsuga canadensis*), and yellow birch (*Betula alleghaniensis*). These forests have low light under their canopies and relatively low understory plant productivity without canopy-opening disturbance (Curtis 1959). While windthrow was not infrequent (Schulte and Mladenoff 2005), this mesic forest ecosystem, the most extensive in Wisconsin and upper Michigan, likely contained the lowest prey biomass per unit area, and thus likely the lowest density of wolves. The pine-dominated sandy outwash plains in the north would have had lower absolute productivity, but because of their drier soils and higher fire frequency, these forests had more open canopies, grading to savanna, and thus greater productivity in the understory.

Northern Minnesota, despite lower productivity based on climate and soil, had higher rates of fire that kept much of the forest in pine and aspen, resulting in higher light levels and more plant productivity within reach of ungulates. Furthermore, the likely higher population of moose relative to deer in northern Minnesota also meant that larger woody browse was available because of the ability of moose to down saplings and feed on bark (Peek 1974).

Given all of this natural variability, it is difficult to conclude what the actual relative prey biomass and therefore wolf population might have been. Jackson (1961) estimated wolf numbers in Wisconsin at the beginning of Euro-American settlement at 20,000–25,000, but this now seems doubtful. A more reasonable estimate based on recent data of wolf densities suggests a possible range of 3,000–5,000 (Wydeven et al. 1995).

8.3 Logging and Post-Logging Era

The era of massive logging, land cover change, and settlement in the northern Great Lakes states occurred from the mid-1800s in the south to the early 1900s in the far north (Mladenoff et al. 1993; Schulte et al. 2007). At the same time, wolves were being extirpated deliberately through bounty programs (Thiel 1993), and indirectly because of prey elimination due to habitat destruction and unregulated hunting (DelGiudice et al., this volume). During the 1900s, habitat for prey improved in reforesting areas following effective fire suppression beginning in the 1930s (Rhemtulla and Mladenoff 2007), and more effective hunting regulation resulted in a gradual resurgence of white-tailed deer especially, and an eventual reestablishment of deer into the southern parts of the region by the 1960s (DelGiudice et al., this volume). In the north, much of this post-fire reforestation was aspen, preferred browse for deer. Since the mid-1900s aspen has declined in Wisconsin and upper Michigan, but not in Minnesota (Schulte et al. 2007).

This change in habitat and prey was little benefit to wolves, since by the 1950s they were extirpated in the region except for wilderness in far northeastern Minnesota (Erb and DonCarlos, this volume). By the 1970s, with wolf protection under the federal Endangered Species Act of 1973, wolves began recolonizing more of northeastern Minnesota, challenging beliefs about the kind of habitat they actually needed. Wolf protection, changing human attitudes, and very high prey (deer) populations complicated examination of this picture, since this was a different landscape in all ways that wolves were re-invading.

8.4 Present Landscape

The land cover composition and pattern of today generally reflects what has existed in the last half century (Fig. 8.1) and the landscape that wolves have recolonized over the past three decades. While land cover alone cannot be read as a habitat map for wolves, it provides key elements. Apart from eradication efforts, human land use activities have drastically altered the 1800s landscape and in fact generally provided better habitat for wolves in the north. In the south, prairie and oak savanna and open woodland have been completely eliminated and replaced by agriculture, with remnant patches of forest. Remnants have become closed canopy forest woodlots due to fire suppression. In the north, the dominant large-statured conifers, pine and hemlock, have also been dramatically reduced and replaced by aspen where slash fires followed the original logging, or younger maple-dominated hardwoods (Rhemtulla et al. 2007). In response, prey, especially deer, have increased dramatically across Wisconsin and the region in the 1900s (DelGiudice et al., this volume).

The regional road network also provides another key layer of data. In the 1980s, knowledge of wolf habitat needs was changing. Legal protection showed that wolves did not need "wilderness" as conceived by humans, but refuge from human contact and its accidental and intentional killing of wolves (Thiel 1985; Mech et al.

1988). Protection and attitude changes reduced intentional killing, and high deer populations may have boosted wolf fecundity. It also became clear that wolves would use roads and snowmobile trails, rather than always avoid them. Clearly, what made up the matrix of favorable wolf habitat was more complex than originally thought at the beginning of wolf recovery.

8.5 Original Great Lakes States Wolf Habitat Model

Thiel (1985) and Mech et al. (1988) identified road density as an important predictor of where wolves would live. By the early 1990s, the first quantitative model of preferred, regional wolf habitat showed that the most important predictor was road density (Mladenoff et al. 1995). Our analysis was based on a logistic regression model of recolonizing wolves in northern Wisconsin:

$$\text{logit}\,(p) = 14.61R - 6.5988$$

where R = road density (km/km^2).

Areas of lower road density (at that time, a threshold of 0.7 km/km^2) were more likely to be occupied by wolves. Since the population was apparently still continuing to recolonize new areas with these characteristics, a more correct interpretation of this pattern would be that wolves were more likely to survive in such areas, and thus tended to establish in these areas first. Occupancy did not have a linear relationship with road density, suggesting that other factors were involved, and that this relationship could continue to change over time (Mladenoff et al. 1997). But mapping these first regional habitat quality classes region-wide and extrapolating estimates of potential wolf numbers (based on the method of Fuller et al. 1992) revealed some starting predictions for potential wolf numbers in the northern Great Lakes states (Mladenoff et al. 1997). At the time, the Wisconsin goal (set in 1989) for downlisting wolves from endangered to threatened status was for a population of 80 wolves, and few thought the population would rise much beyond this level because of the lack of wilderness in the state (Wisconsin DNR 1989). Our estimates suggested that in fact Wisconsin could accommodate 262–662, and upper Michigan 581–1,357 wolves (Mladenoff et al. 1997). In fact, the lower limit of this range has been well surpassed in Wisconsin, with 540 wolves in 2007 (Wydeven et al., this volume). Recent more detailed work in upper Michigan (Potvin et al. 2005) suggested that lower deer densities in high snow areas of upper Michigan may limit wolf numbers in upper Michigan to 10% less than our initial projection. In general, work during the mid-1990s emphasized that managers would likely need to re-assess their estimates and plans for the size of the potential wolf population for the Great Lakes region.

In the late 1990s, we re-assessed our statistical model developed in the early 1990s, using new data from additional recolonizing wolves. This analysis showed that the most favorable classes of habitat with lowest road densities were still being occupied preferentially (Mladenoff et al. 1999). After 2002, wolf numbers approached the lower bounds we had predicted, and were clearly increasing. This was an opportunity to re-examine how our original roads-only-based model described how wolves continued to recolonize. Changes were expected, as many of the most favorable areas were occupied. We did this by recalculating the original roads-based model for the wolf packs occupying Wisconsin in each year from 1979 to 2003. The pattern revealed was a complex one of changes over time and space (D. Mladenoff, unpublished data). Wolves continued preferentially to occupy the most favorable (lowest road density) class, even as this class became increasingly fragmented and reduced. Wolf packs are increasingly occupying less preferred classes, usually by also including portions of the better classes in their territories. Further, they have begun to occupy the lowest quality class, but at very low rates, given that this class constitutes 70% of the landscape (Fig. 8.3; Mladenoff et al., unpublished data). But overall, the original relationship still holds: the wolf population has continued to disproportionately favor the better habitat classes beyond random expectations and availability (Mladenoff et al., unpublished data). As we hypothesized previously, we suspect that the now higher wolf population in better areas can subsidize higher mortality in poorer areas. In such source–sink dynamics (Pulliam 1988), the poorer habitat areas are neither truly biologically suitable in all ways, nor can they be sufficient for a truly sustainable population.

After assessing our original modeling, it is logical to conduct a new model-building exercise based on the current distribution of wolf packs on the current landscape. To do this we used a procedure similar to Mladenoff et al. (1995).

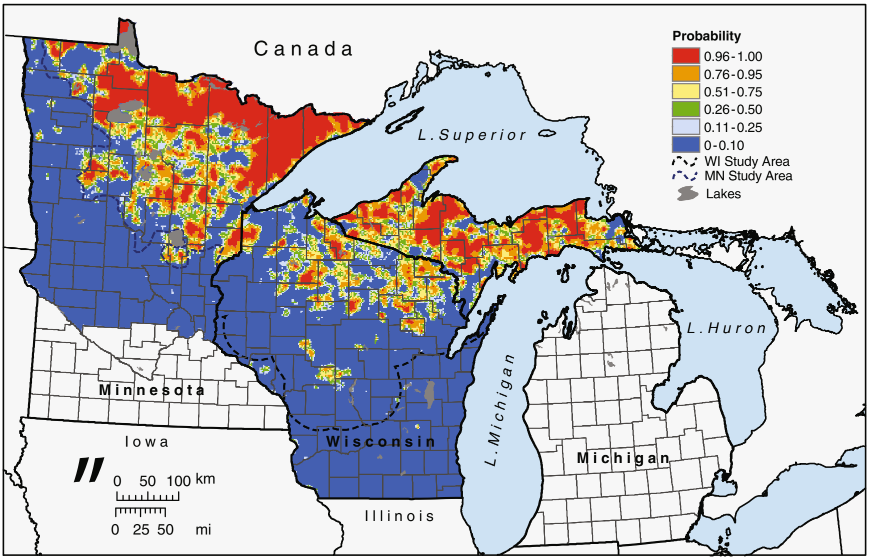

Fig. 8.3 Map of northern Great Lakes states wolf habitat classes from Mladenoff et al. (1995). Map created: Forest Landscape Ecology Laboratory, Department of Forest and Wildlife Ecology, University of Wisconsin-Madison. © David J. Mladenoff (*see Color Plate 3*)

8.6 Development of a New Model of Wolf Occupancy

8.6.1 Study Region

Our study region is located in northern and central Wisconsin and covers two-thirds of the land area of the state, ~97,000 km^2 (Fig. 8.4). When our original model was developed, the study area was defined by the boundaries of the wolf recovery area as established the U.S. Fish and Wildlife Service (1992). Wolves have expanded beyond the original study area so here we used the existing wolf pack locations, buffered by 50 km, to define a new study area. It is possible that the absence of wolf packs beyond this area may not represent less desirable habitat, but may be areas that wolves had not yet reached, and have not had the opportunity to be selected as habitat. In fact, this likely is not true, and is a very conservative assumption. Data suggest that wolves have thoroughly moved throughout the region.

8.6.2 Data Preparation

We acquired the locations of Wisconsin wolf packs for 2007 from the Wisconsin Department of Natural Resources (DNR). Data included polygons of pack territories for 143 packs. Pack territories were mapped using radio-locations from collared wolves or from winter track surveys and reported observations (Wydeven et al. 1995). We then created 143 non-pack areas by randomly locating polygons with an area equal to the mean pack territory size (135 km^2) throughout the study area. These non-pack areas were constrained so they would not overlap with any existing packs or with each other. We randomly selected 95 pack and 95 non-pack areas for use in model development (Fig. 8.4).

We collected and processed several spatial datasets of landscape variables for use as predictor variables in the model including land cover, stream density, and three different metrics based on roads. We used the 2001 National Land Cover Dataset (NLCD), which maps land cover at a 30-m pixel resolution using LANDSAT satellite data (Homer et al 2004). To represent the urban land cover class we used the WISCLAND dataset developed by the Wisconsin DNR (Reese et al 2002). For each pack and non-pack polygon, we calculated the percentage of that area that was occupied by each land cover class. We also aggregated some land cover classes to form additional predictor variables including an agriculture class (crops and pasture) and a forest class (deciduous, mixed, and coniferous forests).

We obtained the 1:1,000,000 scale National Hydrography Dataset (NHD) and created a line shapefile of streams by extracting all streams and artificial paths which did not intersect water bodies (US Geological Survey 1999). We then calculated the density of streams in km/km^2 for each pack and non-pack area. Finally, we used the US Census Bureau 2000 Tiger line data to create a roads layer that included all road categories usable by ordinary cars and trucks but not roads or trails passable only by four-wheel drive vehicles (categories A1, A2, A3, A4, and A6 except A6.5;

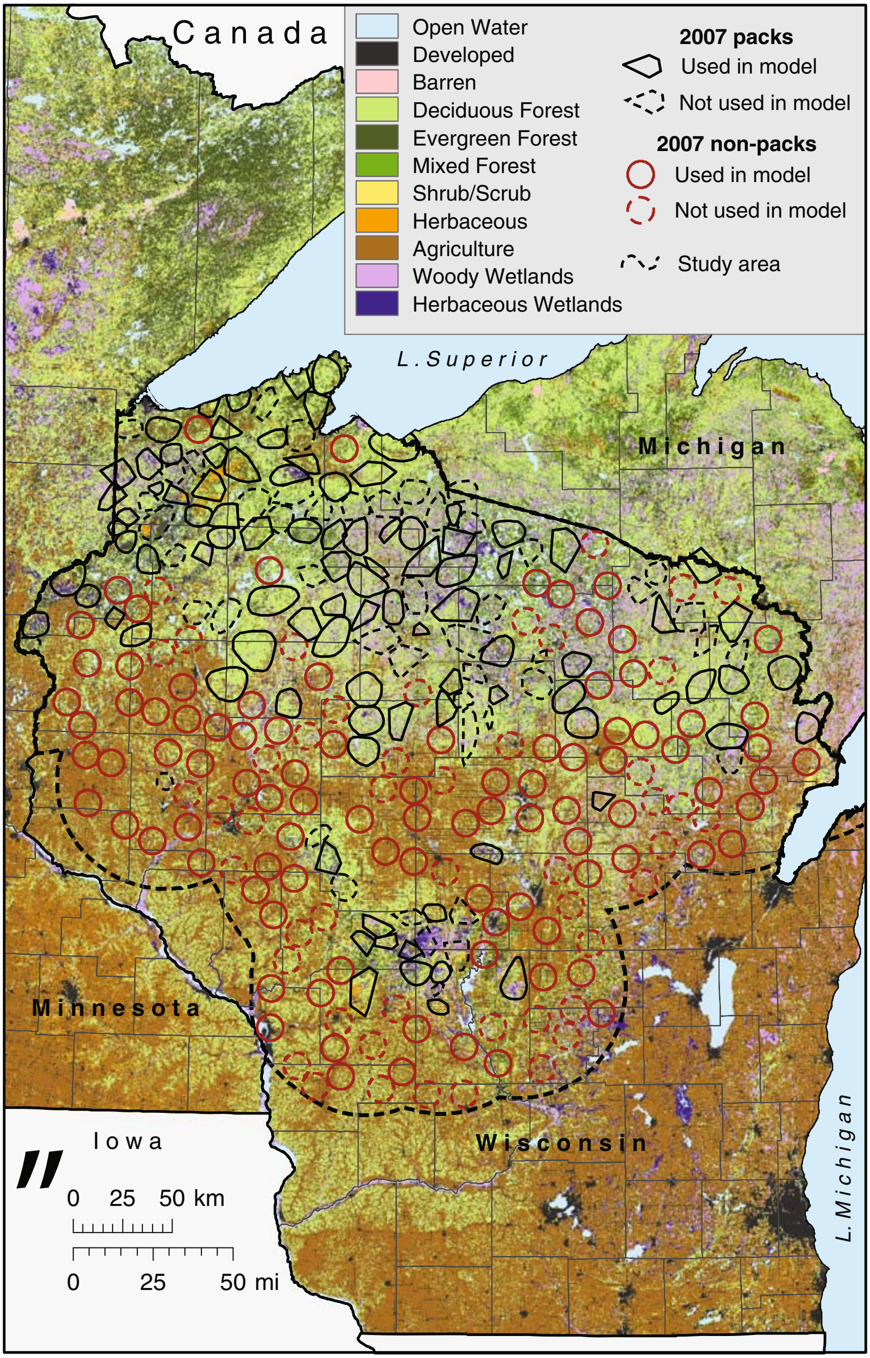

Fig. 8.4 Map of 2006–2007 wolf pack areas and randomly selected non-pack areas used in the analysis for Wisconsin. Map created: Forest Landscape Ecology Laboratory, Department of Forest and Wildlife Ecology, University of Wisconsin-Madison. © David J. Mladenoff (*see Color Plate 4*)

U. S. Census Bureau 2001). From this roads layer, we also created a highway layer that included interstate, USA, and major state and county highways (categories A1 and A2). Similar to streams, we calculated road density and highway density in km/km^2 for each pack and non-pack area. We also calculated the distance, in kilometers, to the nearest highway for each area.

8.6.3 Analysis

We used SAS 9.1 (SAS Institute, Cary, North Carolina) to analyze the data. We calculated summary statistics for the pack and non-pack areas as well as for the Wisconsin study region for each of the landscape variables. We tested for differences in the pack and non-pack areas using the nonparametric Kruskal-Wallis test. A Pearson Correlation Coefficient was calculated between all independent variables to examine multicollinearity. We then used a stepwise logistic regression procedure with all possible variables to find the best set of predictor variables. The resulting model was evaluated based on Akaike's information criterion (AIC, Akaike 1974), the Wald test for maximum likelihood estimates for individual model parameters, and model validation using hold-out pack and non-pack areas. After our initial model fitting, we removed some land cover classes from the stepwise procedure based on analysis of preliminary results and then refit the model. We assessed spatial autocorrelation of the pack locations by calculating semi-variograms of regression residuals, and also by mapping them for visual inspection. Autocorrelation was present but low, and comparison showed that it did not affect the regression results. We therefore used the non-spatial models in the final analysis.

8.7 Results

8.7.1 Landscape Composition

Current landscape composition in pack areas differed between occupied and unoccupied areas (Table 8.1). The greatest differences were in use of agriculture land, which was most negatively correlated with pack locations and constituted 5.3% of pack areas and 41.0% of non-pack areas. The total study area landscape was 27.2% agriculture (Table 8.1, Fig. 8.1). Deciduous forest, by far the largest forest class, was most positively related to pack areas, comprising 49.2% of pack areas and 33.8% of non-pack areas, with the overall study area containing 39.7% deciduous forest. Road density also continued to be negatively associated with packs, with the threshold remaining below 1 km/km^2 for wolf packs. Forested

Table 8.1 Summary of and use and land cover data derived from satellite classification. Means (and standard deviations) for occupied pack territories and randomly located non-pack areas (see modeling description), and for the defined Wisconsin study region overall. Kruskal-Wallis test of differences between pack and non-pack areas are significant at $P < 0.01$ for all classes, except water and shrub/scrub

Variable	Pack territories	Non-pack areas	Study area
Land cover (%)			
Water	2.51 (3.22)	3.18 (4.96)	3.22
Developed	0.03 (0.18)	0.76 (2.84)	0.53
Barren	0.09 (0.15)	0.24 (0.36)	0.26
Deciduous forest	49.24 (17.18)	33.77 (14.46)	39.70
Evergreen forest	8.58 (9.16)	3.98 (5.07)	5.42
Mixed forest	6.84 (7.73)	3.36 (5.67)	5.16
All forest	64.66 (15.25)	41.11 (17.50)	50.28
Shrub/scrub	1.20 (2.20)	1.05 (1.17)	1.20
Herbaceous	3.08 (5.08)	2.66 (2.09)	2.83
Agriculture	5.30 (7.68)	40.98 (23.60)	27.16
Woody wetlands	16.01 (13.05)	7.60 (8.28)	11.00
Emergent wetlands	6.69 (7.64)	2.43 (3.25)	3.51
Road density (km/km^2)	0.93 (0.35)	1.42 (0.48)	1.31

wetlands and forest cover overall also were major land cover classes that showed a strong positive association with wolf pack territories (Table 8.1).

Correlations among the variables reflected these positive and negative associations, with the forest variable being most strongly negatively correlated with agriculture and roads, and positive correlations within the variables of these two groups (Table 8.2).

8.7.2 New Model

Our new, current model of wolf occupancy revealed how behavior has changed since the early 1990s. In this model (Figs. 8.5 and 8.6), the broader extent of wolf occupancy is reflected in a change in the significant variables in the model. While road density was the only significant variable in the 1995 model, in the current analysis model results were more varied. For direct comparison with the 1995 model, we calculated a roads-only model from the current data. This model remained a significant predictor of wolf presence, though it was not among the best models found.

More comprehensive analyses (including other predictors) resulted in several models of nearly identical AIC values and validation, but varying in their interpretability. One included road density, mixed forest, and agriculture, with all having negative coefficients for wolf prediction. This result was unexpected for mixed forest, which by itself was highly correlated with wolf presence (Table 8.1). The negative coefficients for roads and agriculture are expected, because these are variables that indicate high human presence on the landscape (Mladenoff et al. 1995). This complex

Table 8.2 Pearson correlation matrix for land use and land cover variables within the occupied pack areas for wolf habitat in Wisconsin

	Open water	Developed	Barren	Deciduous forest	Conifer forest	Mixed forest	All forest	Agriculture	Shrub/scrub	Herbaceous	Woody wetland	Herbaceous wetland	Road density	Highway density	Distance to highway	Stream density
Open water	1.00	0.21	0.09	–0.21	0.18	0.38	0.02	–0.28	–0.06	–0.04	0.13	0.09	0.28	–0.07	–0.04	–0.06
Developed	0.21	1.00	0.21	–0.16	–0.04	–0.07	–0.18	0.07	–0.04	0.10	–0.09	–0.05	0.65	0.23	–0.13	0.00
Barren	0.09	0.21	1.00	–0.32	–0.11	0.01	–0.32	0.32	–0.07	0.03	–0.15	–0.13	0.32	0.33	–0.27	–0.11
Deciduous forest	–0.21	–0.16	–0.32	1.00	0.12	–0.21	0.85	–0.54	–0.08	–0.10	–0.12	–0.07	–0.42	–0.25	0.24	0.22
Conifer forest	0.18	–0.04	–0.11	0.12	1.00	0.07	0.51	–0.42	–0.07	0.28	–0.20	0.12	0.14	–0.12	0.15	0.02
Mixed forest	0.38	–0.07	0.01	–0.21	0.07	1.00	0.19	–0.46	0.00	–0.22	0.62	–0.03	–0.10	–0.13	0.02	0.20
All forest	0.02	–0.18	–0.32	0.85	0.51	0.19	1.00	–0.79	–0.09	–0.05	0.03	–0.03	–0.35	–0.31	0.28	0.27
Agriculture	–0.28	0.07	0.32	–0.54	–0.42	–0.46	–0.79	1.00	–0.07	–0.04	–0.49	–0.31	0.40	0.38	–0.35	–0.29
Shrub/scrub	–0.06	–0.04	–0.07	–0.08	–0.07	0.00	–0.09	–0.07	1.00	0.39	0.02	0.08	–0.06	–0.04	0.00	–0.02
Herbaceous	–0.04	0.10	0.03	–0.10	0.28	–0.22	–0.05	–0.04	0.39	1.00	–0.29	0.15	0.22	0.04	–0.06	–0.21
Woody wetland	0.13	–0.09	–0.15	–0.12	–0.20	0.62	0.03	–0.49	0.02	–0.29	1.00	0.10	–0.40	–0.24	0.19	0.23
Herbaceous wetland	0.09	–0.05	–0.13	–0.07	0.12	–0.03	–0.03	–0.31	0.08	0.15	0.10	1.00	–0.26	–0.12	0.29	0.01
Road density	0.28	0.65	0.32	–0.42	0.14	–0.10	–0.35	0.40	–0.06	0.22	–0.40	–0.26	1.00	0.36	–0.31	–0.18
Highway density	–0.07	0.23	0.33	–0.25	–0.12	–0.13	–0.31	0.38	–0.04	0.04	–0.24	–0.12	0.36	1.00	–0.42	–0.10
Distance to highway	–0.04	–0.13	–0.27	0.24	0.15	0.02	0.28	–0.35	0.00	–0.06	0.19	0.29	–0.31	–0.42	1.00	0.06
Stream density	–0.06	0.00	–0.11	0.22	0.02	0.20	0.27	–0.29	–0.02	–0.21	0.23	0.01	–0.18	–0.10	0.06	1.00

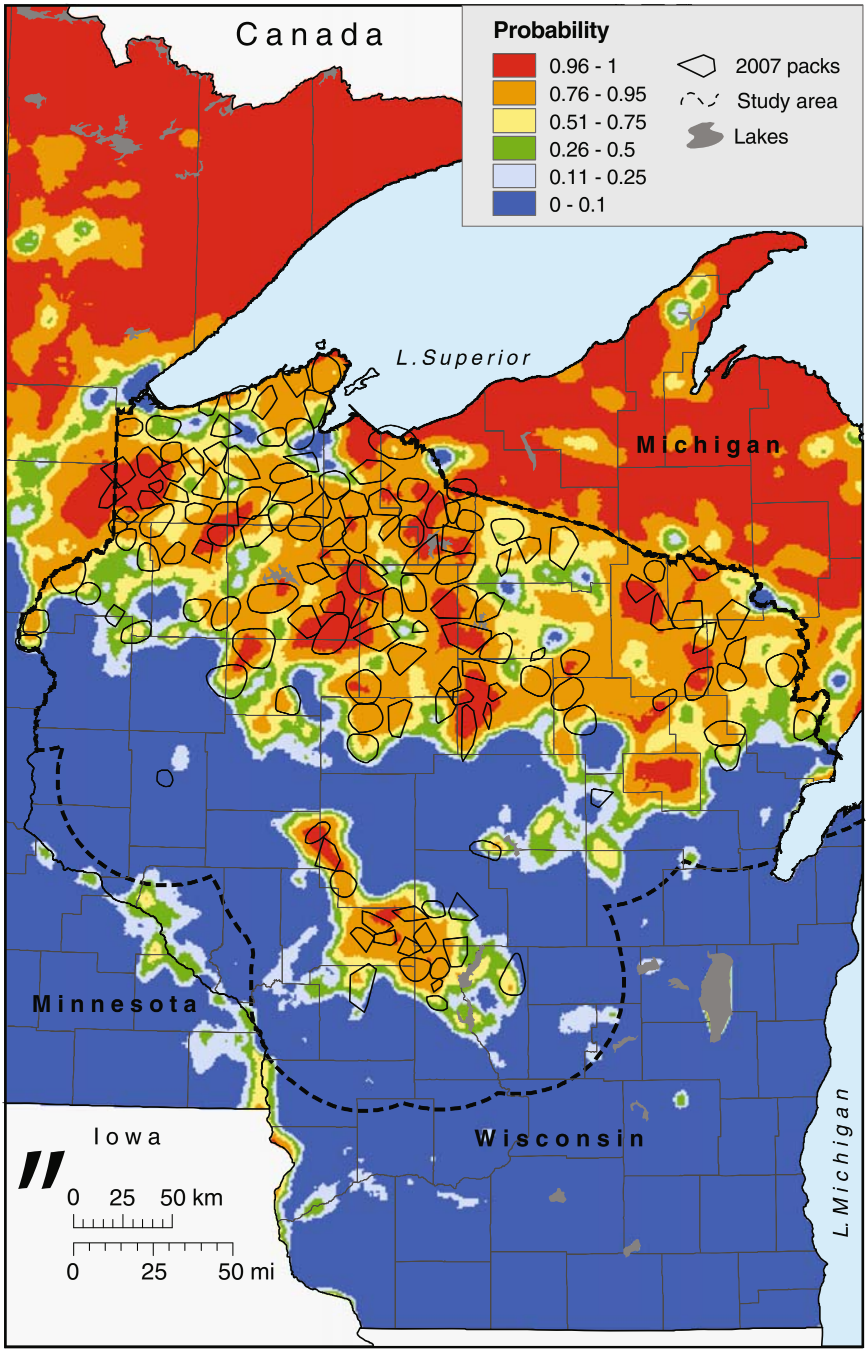

Fig. 8.5 Map of wolf habitat probability classes in Wisconsin based on new analysis and wolf pack occupancy in winter 2006–2007. Map created: Forest Landscape Ecology Laboratory, Department of Forest and Wildlife Ecology, University of Wisconsin-Madison. © David J. Mladenoff (*see Color Plate 5*)

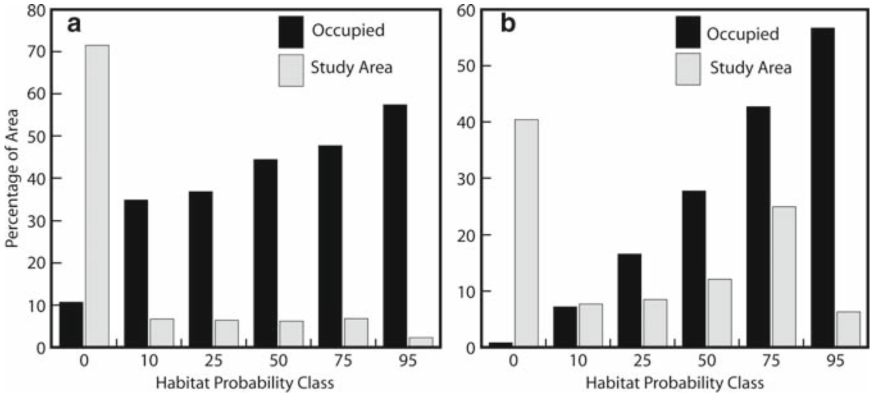

Fig. 8.6 Percentage of habitat area classes in the Wisconsin study area and occupied percentage for **(a)** the new model (map Fig. 8.7) and **(b)** the original 1995 model (map Fig. 8.3).

model result may occur because mixed forest is a small forest class in the landscape, and co-occurs with a relatively high road density area in the north central lakes area (Table 8.1, Fig. 8.1). Hence, complex interactions among variables resulted in a low but negative coefficient for mixed forest. To address this, we merged the three forest classes into one forest class and repeated the model-building exercise. The most interpretable model with equally high AIC and validation to the roads-ag-mixed forest model included roads and agriculture only:

$$\text{Logit}\,(p) = 5.0018 - 11.7095A - 2.5655R,$$

where p is the probability of occurrence of a wolf pack, A is agricultural land use (%), and R is road density (km/km^2, Table 8.3).

Validation was nearly identical for the models, with a cut level at the ≥70% probability level resulting in 81.6% correct classification, with 40 pack and 40 non-pack areas (80) correctly classified, and eight each (16) misclassified. In this current model, much more of the landscape is in high occupancy classes (>75% probability, Fig. 8.6) compared to the 1995 roads-only model (Fig. 8.4). This reflects the fact that a high proportion of the forested landscape (Fig. 8.1) is now occupied, but wolves typically avoided agriculturally dominated areas. The highest probability habitat (>95% probability) remained strongly influenced by road density, as shown by the conjunction of this class in the model maps from the two periods (Figs. 8.3 and 8.5).

8.8 Discussion

Our analysis of wolf habitat and information on current distribution of wolves demonstrates that wolf packs are able to occupy areas well beyond wilderness across the Great Lakes region. Given adequate protection, wolves are able to occupy most

Table 8.3 Logistic regression model statistics for final model

Parameter	DF[a]	Estimate	SE[b]	Wald X^2	*P*
Intercept	1	5.00	0.95	27.49	<0.01
Road density	1	–2.57	0.74	12.07	<0.01
Agriculture	1	–11.71	1.99	34.75	<0.01

[a]Degrees of freedom
[b]Standard error

large tracts of forest and other wildland habitat. Similar selection for forest and wildlands, while avoiding anthropogenic features such as agricultural land, roads, and areas of high human density, was demonstrated through similar habitat modeling in the northern Rocky Mountains of the USA (Oakleaf et al. 2006), across Poland (Jędrzejewski et al. 2004, 2005), and in Italy (Corsi et al. 1999).

Mech (2006) questioned the value of this type of modeling for predicting wolf distributions in Wisconsin, though his earlier analysis also showed the value of using road densities to describe wolf distribution in Minnesota. Mech's critique (2006) of our modeling efforts was not based on sound analysis (Mladenoff et al. 2006). Our analysis demonstrated the usefulness of our earlier model for predicting initial colonization and direction of habitat occupancy, and it continues to be robust. Our new model probably more fully describes the extent of future pack occurrences under current human-caused mortality regimes. If human-caused mortality rates change in the future, areas of potential wolf habitat would likely change as well.

We suggest that our new model and the 1995 model reflect differences not only due to the changed number and location of wolves, but that the models also mean very different things and should be viewed and used in different ways. This is particularly true if we extrapolate the new Wisconsin model to the three-state region (Fig. 8.7).

Our original (Mladenoff et al. 1995) model reflected a small, new, colonizing wolf population in northern Wisconsin and showed how wolves were occupying the landscape in a very selective way. This pattern changed gradually over intervening years, as wolf numbers increased from 25 to >540 (Wydeven et al., this volume). Over this time, our original model was able to continue to identify the most preferred habitat (Mladenoff et al. 1999; Mladenoff et al., unpublished data). Though its predictability gradually declined in the most recent half decade, it remained a significant predictor of wolf distribution, and road density remains a key secondary variable in the new model, especially for indicating the highest quality habitat areas.

Our new model reflects different information. This model reflects current habitat occupancy, but likely under very different dynamics, not merely higher and more broadly dispersed wolf numbers. We suggest that our current model reflects the result of strong source–sink dynamics (Pulliam 1988) as suggested by our earlier work (Mladenoff et al. 1995, 1999, 2006). We hypothesize that our new model shows wolf occupancy under the influence of strong fecundity and available dispersers from core, more reliable habitat that is smaller and more fragmented in northern Wisconsin, and is more abundant in Minnesota and upper Michigan (Table 8.4, Figs. 8.3 and 8.7). These core areas are shown in red in the original model (Fig. 8.3),

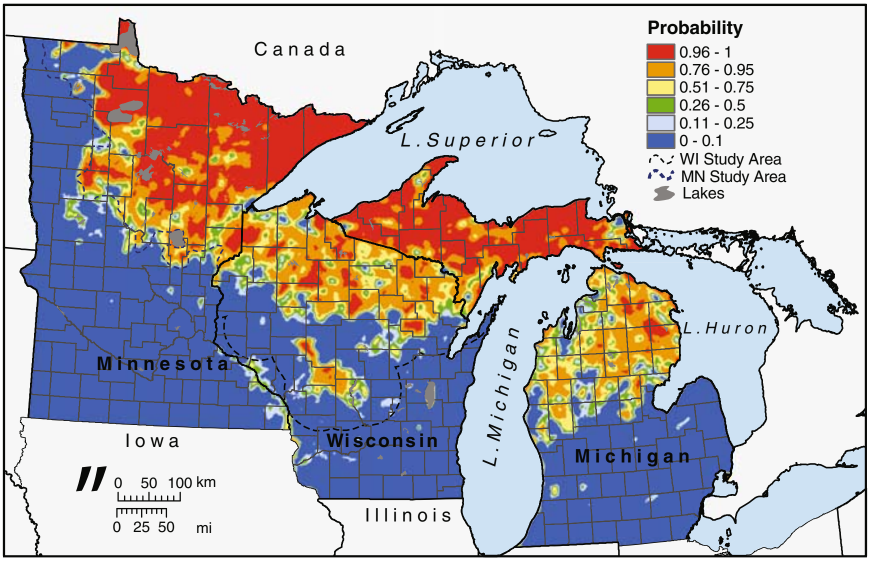

Fig. 8.7 Mapped extrapolation of the new Wisconsin model to the three states in the Great Lakes region. Map created: Forest Landscape Ecology Laboratory, Department of Forest and Wildlife Ecology, University of Wisconsin-Madison. © David J. Mladenoff (*see Color Plate 1*)

Table 8.4 Percentage of probability classes for the new wolf habitat model extrapolated to the three-state area

	Michigan				Minnesota		Wisconsin	
	Lower Peninsula		Upper Peninsula		Wolf range		Study area	
Probability class (*p*)	Area (km^2)	(%)	Area (km^2)	(%)	Area (km^2)	(%)	Area (km^2)	(%)
≥0.95	2,143	2.0	29,818	68.9	52,927	42.2	6,121	6.3
0.75–0.94	15,513	14.5	9,775	22.6	20,401	16.3	24,161	25.0
0.50–0.74	10,609	9.9	1,826	4.2	7,280	5.8	11,735	12.1
0.25–0.49	7,246	6.8	1,082	2.5	5,812	4.6	8,220	8.5
0.10–0.24	5,645	5.3	442	1.0	5,033	4.0	7,432	7.7
<0.10	65,793	61.5	309	0.7	33,972	27.1	39,147	40.4
Total	106,949	100	43,252	100	125,425	100	96,816	100

and served as secure source and population expansion areas during the colonization years of the 1980s through early 2000s. Subsequent continued colonization into larger portions of the map (class ≥75% probability, Figs. 8.6 and 8.7) may have been dependent on those secure source areas. This may be why avoidance of agriculture (essentially, high-human-contact areas) is now the key variable in the model. Road density reflected similar information, but areas of moderate road density also occur with some areas of high forest cover that remain important in the new model.

This interpretation further suggests that our new model should not be seen as a map of favorable habitat that the 1995 model reflects. In particular, the new model should not be interpreted in that way in disjunct areas such as lower Michigan (Gehring and Potter 2005), where there is only limited access to the nearby source populations (Fig. 8.7). Our new map may better reflect potential habitat in upper Michigan and Minnesota, especially in the large expanses of high-quality source habitat (Fig. 8.7). It is unlikely that wolf occupancy will change again in the near future as significantly as it has in the last decade. There is evidence in Minnesota that expansion has slowed, possibly reflecting an equilibrium between the source population colonizers and increasing mortality in increasingly less favorable areas (Erb and DonCarlos, this volume), and population growth has declined in Wisconsin and Michigan, possibly due to a similar mechanism (Van Deelen, this volume). Also, wolf control has increased gradually as the growing population has spread into areas with more human contact. Federal delisting and more flexible controls applied by the states may further cause wolf range to stabilize at the agricultural–forest fringe. These factors may make it difficult to separate causes of any observed reduction in expansion of occupied wolf range.

Our earlier model of favorable habitat (Mladenoff et al 1995), and projected wolf populations (Mladenoff et al. 1997), were used extensively for wolf management planning in Wisconsin (Wisconsin DNR 1999). Although wolf pack distribution has extended beyond the original areas predicted to become occupied by packs, the models were used for creating management zones, establishing population goals, planning surveys, and planning for future population expansion. Our new model should also provide useful information for future habitat and population management for wolves in the Great Lakes region and will generate important research opportunities for understanding eventual limits to wolf population growth and range expansion.

8.9 Summary

Occupied wolf habitat in the Great Lakes region declined from region-wide at the time of Euro-American settlement to remote wilderness areas by the mid-1900s. By the time wolves were listed as a federal endangered species in 1974, they were perceived mainly as a wilderness species. Through protection of the Endangered Species Act, wolves were able to expand to forests and other wildland areas across the states of Minnesota, Michigan, and Wisconsin. Our modeling of wolf habitat in the 1990s showed that road densities were the best predictors of areas of initial colonization and direction of population spread. As core habitat areas have been mostly occupied and wolves have spread across much of the region's large expanses of forest, deriving a new model has shown both the changes and persistence of wolf preferences on the landscape. In our new model, the best predictors of wolf habitat are lack of agricultural land and low road density. Our mapping of wolf habitat shows extensive areas of suitable wolf habitat across the region, but core habitat is more fragmented in Wisconsin than in northern Minnesota and upper

Michigan. We anticipate that potential for further range expansion will be limited in the region, except possibly if wolves are able to expand to lower Michigan (Gehring and Potter 2005).

Conservationists should consider carefully our two modeling results in assessing future wolf population dynamics and management. Most importantly, our new model should be seen as more descriptive of wolf occupancy, and more dependent on complex population dynamics, than our simpler 1995 model. We suggest that the 1995 model is the best conservative indicator of preferred, most-critical habitat, especially in Wisconsin where both models show that core habitat remains more limited. This suggests that maintaining population security in Wisconsin will continue to require habitat management that limits road development and maintains areas of low road density on public forest lands, so that these areas can continue to function as core habitat. Long-term viability of Wisconsin wolves will likely require maintaining connectivity to the abundant high-quality source areas in Minnesota and upper Michigan.

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